Breathing Techniques and Breath-Regulation Strategies in Sport and Performance Contexts: A Scoping Review

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Abstract

Background: Breathing techniques are widely used in sport and performance settings, but the evidence spans many techniques, populations, outcomes, and reporting styles. **Objective:** This scoping review maps evidence on non-device breathing and breath-regulation strategies, integrating bibliometric, participant, intervention, efficacy, feasibility, and safety data. **Methods:** The review followed PRISMA-ScR principles. Sixty-five reports were charted and synthesized descriptively; exploratory quantitative syntheses were conducted when outcome data were extractable. Efficacy effects were standardized so that positive Hedges' g values favored the breathing intervention. Safety and dropout outcomes were summarized as risk ratios when extractable. **Results:** The review included 65 reports from 2000–2026. Most reports were experimental (58/65, 89.2%), with crossover designs forming the largest detailed group (28/65, 43.1%). Sample size was extractable for 47 reports, totaling 1,799 participants. The most frequent breathing categories were hyperventilation (16/65, 24.6%), hypoventilation (14/65, 21.5%), paced breathing (13/65, 20.0%), and breath holding/apnea (10/65, 15.4%). Exploratory pooling showed small and uncertain effects for performance outcomes ($g = 0.09$, 95% CI -0.16 to 0.34) and physiological outcomes ($g = 0.21$, 95% CI -0.22 to 0.64). Safety and dropout risk-ratio syntheses did not show clear overall differences. **Conclusions:** The field is active and mechanistically plausible, but methodologically uneven. Current evidence supports clearer intervention taxonomy, rigorous testing of sport-derived breathing routines, standardized outcome selection, and routine reporting of feasibility and adverse events.

Key takeaways

- The review includes 65 reports covering breathing techniques across sport, exercise, and performance contexts.
- Breathing interventions should not be treated as one uniform "breathwork" category.
- Pooled efficacy signals are small, uncertain, and best treated as hypothesis-generating.
- Feasibility and safety reporting are still not strong enough for confident implementation guidance.
- Future studies should test empirically developed sport breathing routines with clear protocols, credible comparators, and complete reporting.

Keywords: scoping review; PRISMA-ScR; breathing training; breathwork; hyperventilation; hypoventilation; paced breathing; sport performance; feasibility; safety

1 Introduction

Breathing is automatic, but several of its parameters can be deliberately regulated. Respiratory frequency, depth, inhalation-exhalation ratio, breath-hold timing, route of breathing, and lung volume can all be modified intentionally. This makes breathing unusual among performance-relevant physiological functions: it is continuously available, requires little equipment, and can be applied before exercise, during exertion, during recovery, or as part of repeated training. In sport, athletes and practitioners use breath-regulation strategies to influence arousal, anxiety, oxygen saturation, carbon dioxide tolerance, acid-base balance, perceived exertion, attentional control, respiratory mechanics, and recovery.

The challenge is that "breathwork" is not one intervention. Hyperventilation, hypoventilation, paced breathing, breath holding, pranayama, diaphragmatic breathing, hook breathing, and breathing-pattern retraining have different mechanisms and risks. Some techniques aim to increase activation; others aim to down-regulate stress or support recovery. Some are single-session pre-performance routines; others are repeated training exposures. Some resemble sport-specific hypoxic or hypercapnic training, while others are better understood as psychophysiological regulation tools.

Previous synthesis has shown why this distinction matters. In a systematic review and meta-analysis of breathing techniques for physical sport performance, Laborde and colleagues separated slow-paced breathing, fast-paced breathing, voluntary hyperventilation, breath holding, and nostril-breathing approaches, and reported more favorable signals for some longer-term slow-paced breathing and breath-holding interventions than for short-term applications, while substantial heterogeneity and risk-of-bias concerns remained (Laborde et al., 2024). Their review also highlighted a practical boundary that is important for sport: breathing techniques can often be performed in the field without devices that directly alter respiratory load or inspired gas composition, although biofeedback may be used to monitor or guide execution rather than to produce the respiratory stimulus itself.

This heterogeneity makes a scoping review appropriate. The purpose is not only to ask whether breathing techniques work, but to map what has been studied, how techniques have been categorized, which sport contexts and populations have been examined, which outcomes have been used, and where feasibility and safety reporting remain incomplete. The review questions were: (1) which breathing-technique families have been studied in sport and performance contexts; (2) which populations, designs, outcomes, and intervention doses characterize the evidence base; and (3) what can be learned from extractable efficacy, feasibility, and safety data without treating heterogeneous techniques as a single intervention.

2 Methods

2.1 Design and reporting

This scoping review was designed to map the breadth, structure, and reporting completeness of evidence on breathing techniques in sport and performance contexts. The manuscript is reported according to PRISMA-ScR principles. The review maps the available evidence rather than claiming a single definitive intervention effect. The unit of descriptive synthesis was the report or study row represented in the extraction table. Quantitative syntheses were exploratory adjuncts to the evidence map and used only extractable effect rows from the accompanying meta-analysis dataset.

2.2 Protocol and registration

No registered protocol was identified in the project files available for this manuscript draft.

2.3 Eligibility criteria

Eligibility was defined using a Population–Concept–Context structure. Eligible populations included athletes, sport participants, physically active participants, and healthy participants completing performance-relevant exercise or psychophysiological tasks. The concept was a breathing technique, breath-regulation routine, or breathing-related exposure that could be performed without a device that directly altered respiratory load or inspired gas composition. Eligible concepts included hyperventilation, hypoventilation, paced or slow breathing, breath holding/apnea, pranayama or yogic breathing, diaphragmatic breathing, hook breathing, Valsalva-type strategies, and related approaches. Eligible contexts included sport training, laboratory exercise tasks, sport-specific testing, recovery, and performance-relevant psychophysiological outcomes.

Source types included primary empirical reports with extractable study-level information; reviews were used for citation searching but were not included as primary evidence rows.

2.4 Information sources and search strategy

Searches were run in PubMed, Scopus, and SPORTDiscus. The database-specific search strings are reported in the appendix. The searches combined terms for healthy or athletic populations, breathing techniques, sport or performance outcomes, and exclusions for reviews, respiratory muscle training, pulmonary disease, and major clinical respiratory contexts. Additional reports were identified through citation searching, backtracking from relevant reviews, and manual reconciliation during full-text assessment.

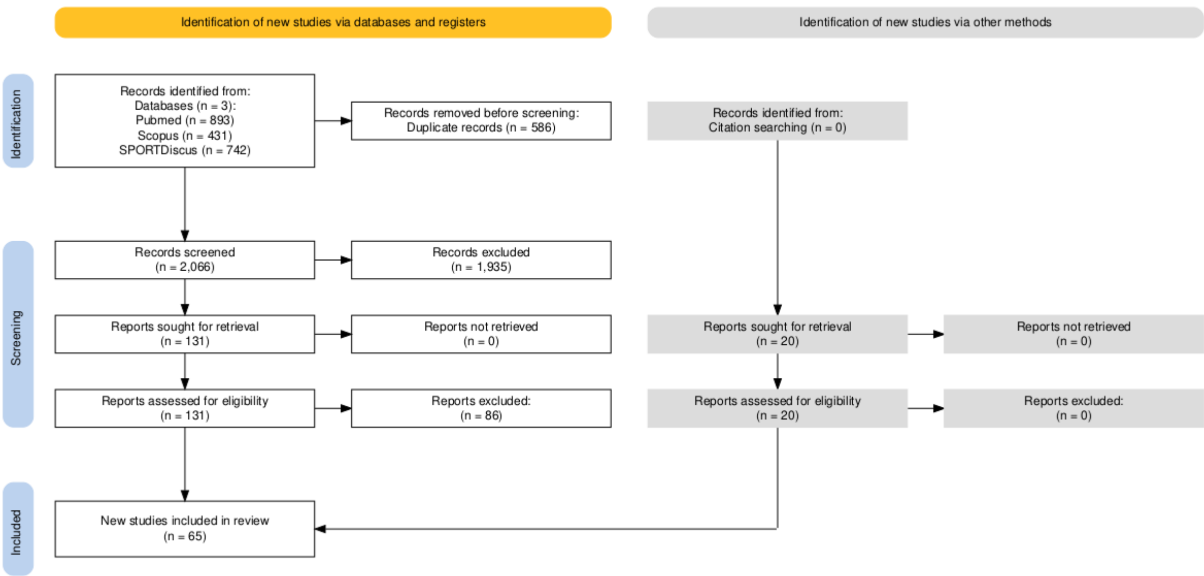


Figure 1: PRISMA flow diagram for study identification, screening, eligibility assessment, and inclusion.

Figure 1 shows that database searching identified 2,652 records before duplicate removal and 2,066 records after duplicate removal. One hundred thirty-one reports were assessed for eligibility from database searching. A further 20 reports were assessed from citation searching and other methods, including backtracking and manual additions identified during full-text reconciliation. In total, 65 reports were included.

2.5 Selection process

Records were screened at title/abstract level, followed by full-text assessment of potentially eligible reports. Full texts were assessed against the population, concept, context, and source-type criteria described above. Disagreements and uncertain cases were resolved during eligibility reconciliation before final extraction.

2.6 Data charting and quality control

Data were charted for bibliometry, study design, sport context, athlete level, sample size, age, sex distribution, baseline characteristics, breathing category, intervention dose, comparator, efficacy outcomes, uncertainty, feasibility, dropout, adverse events, and safety. Missing or unreported fields were retained as `NR` so that denominators and reporting gaps remained explicit. Data-quality checks flagged missing sample-size fields, unclear breathing categories, and sparse feasibility or safety reporting for manual review.

2.7 Outcome grouping and synthesis

Descriptive synthesis summarized publication year, study design, participant characteristics, sport context, breathing category, intervention duration, intervention frequency, feasibility, and safety. Outcomes were grouped into performance, physiological, feasibility, and safety domains. Exploratory meta-analysis used Hedges' g for efficacy outcomes when group-level data were extractable. Effect direction was standardized so that positive values favored the breathing intervention; lower-is-better outcomes such as time, reaction time, heart rate, RPE, dyspnea, fatigue, lactate, and decrement scores were reversed. Safety and dropout were synthesized as risk ratios where event data were available. Very imprecise safety estimates were retained in extraction records but excluded from displayed forest plots when the interval made the figure unreadable.

2.8 Feasibility and safety definitions

Feasibility fields included adherence, dropout, acceptability when available, and reasons for discontinuation. Safety fields included adverse-event reporting status, adverse-event descriptions, and event counts when extractable. Absence of reported adverse events was not interpreted as evidence that no events occurred unless the source explicitly stated this.

2.9 Critical appraisal

Formal risk-of-bias appraisal was not the primary objective of this scoping review. Instead, the synthesis emphasizes reporting completeness, design heterogeneity, missing feasibility fields, and uncertainty around quantitative estimates. This approach is appropriate for mapping a heterogeneous field, but it limits causal inference and prevents strong claims about efficacy.

3 Results

The Results are organized to move from source selection and cohort description to the structure of the evidence map, then to intervention categories, efficacy signals, and feasibility and safety. This order reflects the purpose of a scoping review: first to define what the evidence base contains, and only then to interpret what the extractable outcomes suggest.

3.1 Study and participant characteristics

Table 1: Aggregate study and participant characteristics.

Characteristic	Summary
Reports in extraction corpus	65 reports
Publication years	2000–2026
Reports with extractable total sample size	47 reports
Aggregate reported sample size	1,799 participants
Age reporting	25 reports; weighted mean 28.1 years; pooled SD 10.1 years
Sex reporting	1,348 participants with sex data; estimated male proportion 59.8%
Athlete level	Competitive: 33; physically active: 14; elite: 6; recreationally trained: 3; healthy non-athlete: 2; not reported/blank: 7
Design profile	Experimental: 58 (89.2%); observational: 5 (7.7%); unclear: 2 (3.1%)

Table 1 summarizes the included corpus. The evidence base is dominated by experimental work, but this should not be read as uniform trial evidence: many reports use crossover, acute, or otherwise heterogeneous designs. The aggregate reported sample size was 1,799 participants, although sample size was not extractable in 18 rows, which limits precision when describing the population base. The full study-level extraction table is provided as Supplementary Table S1.

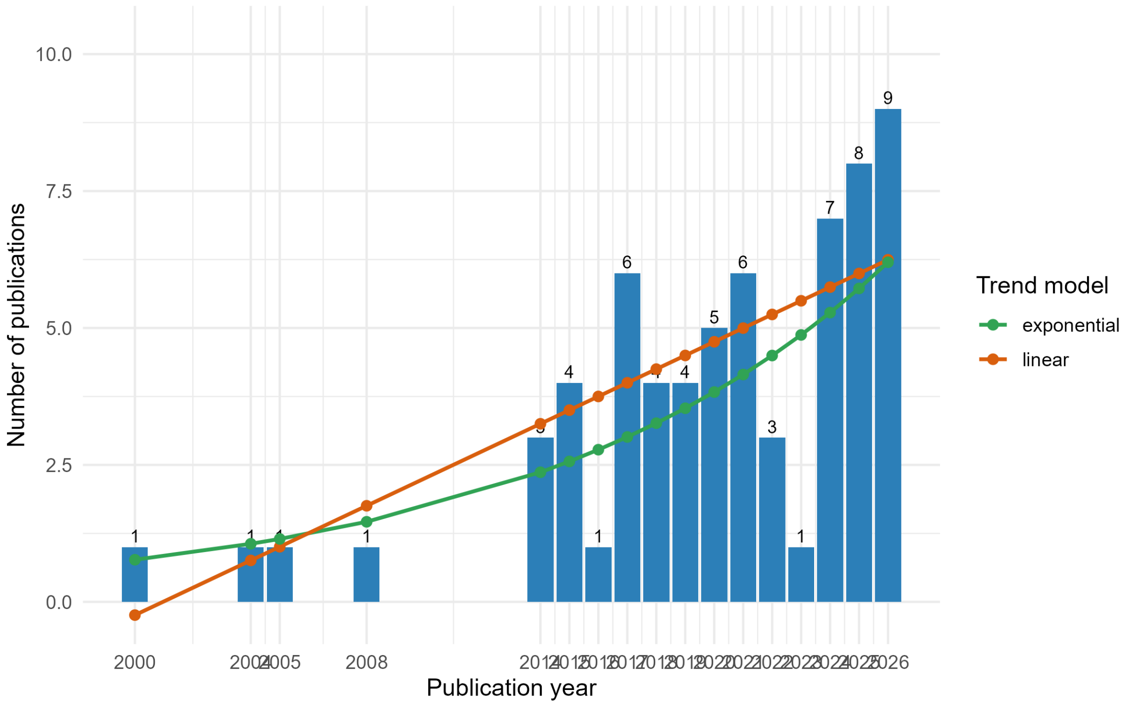


Figure 2: Annual publication counts and fitted trend lines.

Figure 2 indicates that research activity increased after 2014 and was especially dense in the most recent years. This growth supports the need for mapping, but publication volume should not be mistaken for methodological maturity.

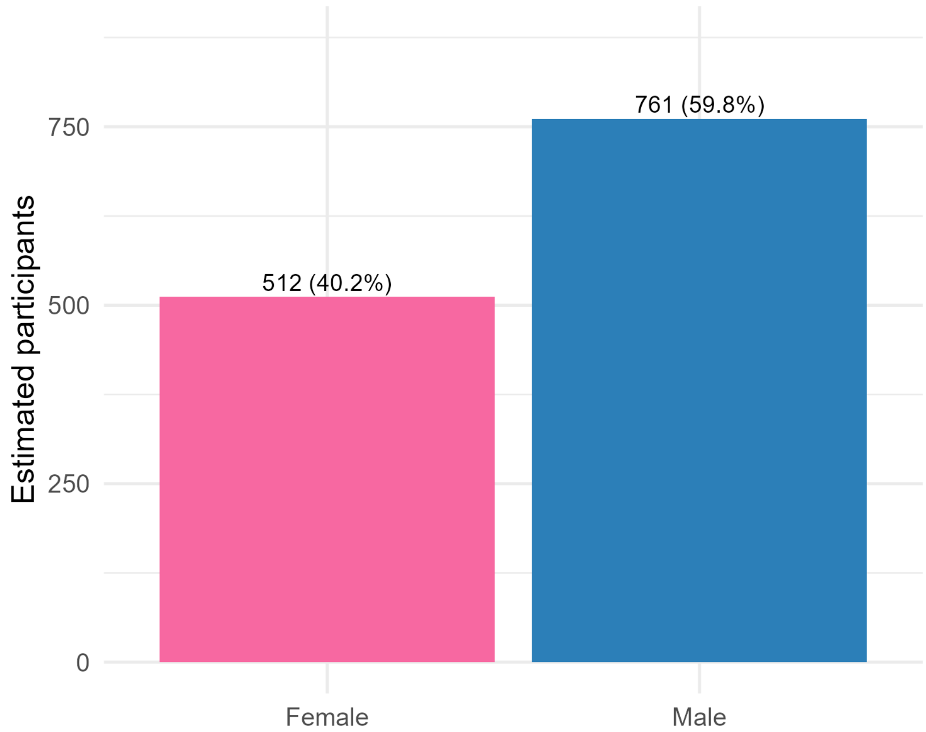


Figure 3: Estimated sex distribution among studies with extractable sex data.

Figure 3 indicates a male-predominant participant pool among reports with sex data. This imbalance matters for generalizability because technique-specific responses and implementation contexts may not transfer cleanly across sex-balanced or female athlete populations.

3.2 Evidence-map structure

Table 2: Evidence-map summary.

Domain	Main result
Detailed study designs	Crossover: 28 (43.1%); other/unclear experimental: 25 (38.5%); single-group/observational: 5 (7.7%); before-after: 3 (4.6%); acute experimental: 2 (3.1%); unclear: 2 (3.1%)
Most represented sport contexts	Cycling: 9 (13.8%); swimming: 9 (13.8%); unclear/not specified: 7 (10.8%); running: 3 (4.6%); burpees, freediving, HIIT, and soccer: 2 each (3.1%)
Breathing categories	Hyperventilation: 16 (24.6%); hypoventilation: 14 (21.5%); paced breathing: 13 (20.0%); breath holding/apnea: 10 (15.4%); other/unclear: 7 (10.8%); diaphragmatic: 2 (3.1%); pranayama/yogic: 2 (3.1%); hook breathing: 1 (1.5%)
Dose duration	33 parseable reports; median 1 day; IQR 1–28 days
Session frequency	23 parseable reports; median 1 session/week; IQR 1–2.5 sessions/week

Table 2 summarizes the key structural findings. The corpus is broad but uneven, with several technique and sport categories represented by small clusters. The map therefore supports technique-specific interpretation rather than a single pooled claim about "breathwork".

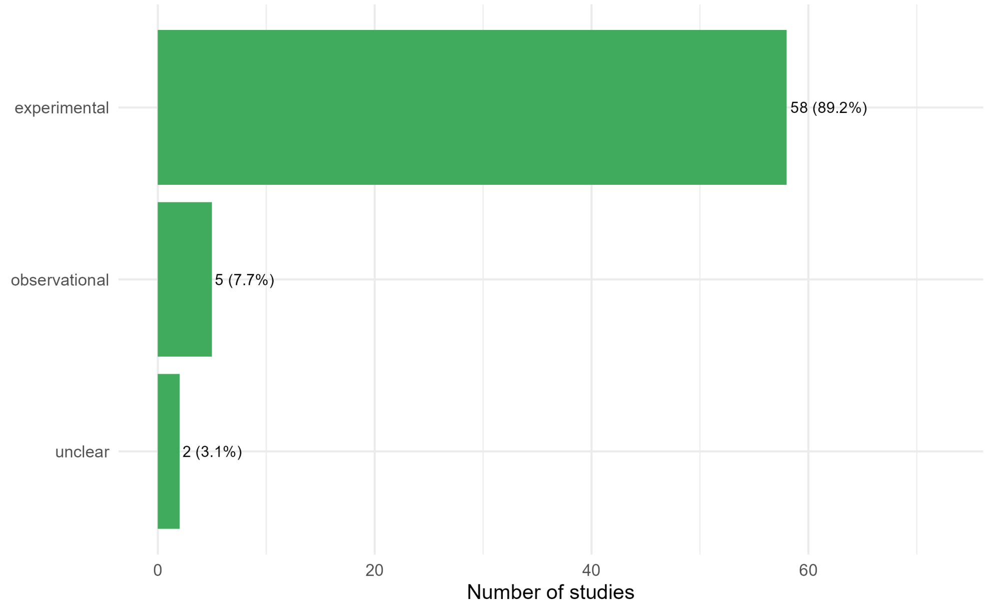


Figure 4: Broad study-design distribution.

Figure 4 shows that experimental reports dominate the corpus. This supports intervention mapping, but it does not by itself imply high certainty because many studies remain acute, small, or heterogeneous.

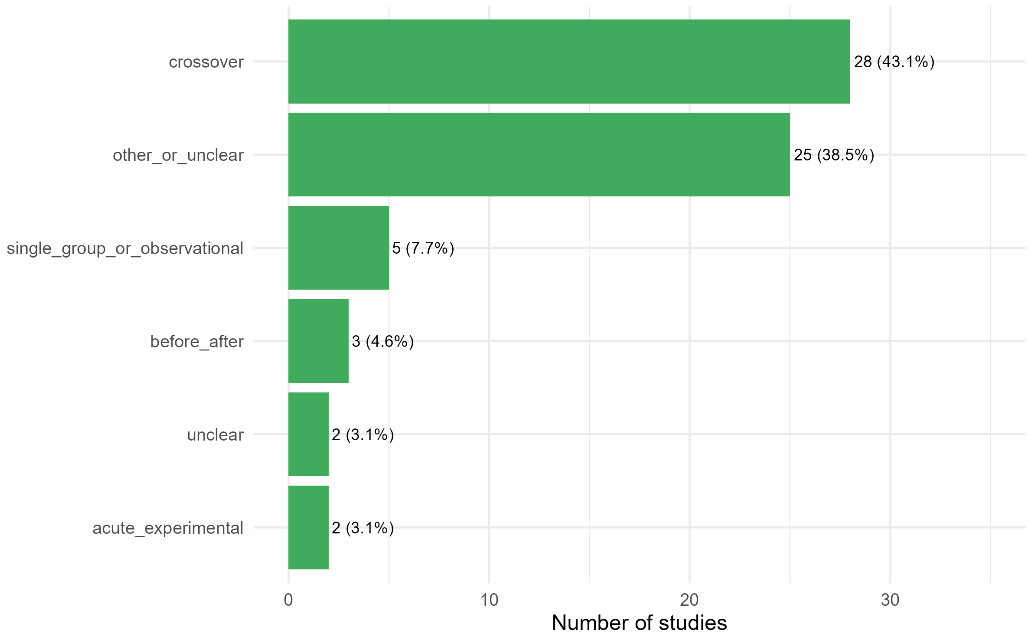


Figure 5: Detailed study-design distribution.

Figure 5 clarifies the design imbalance: crossover and other short-form experimental designs are common, whereas durable parallel-group training trials are less prominent. This affects how confidently the field can move from acute responses to practical recommendations.

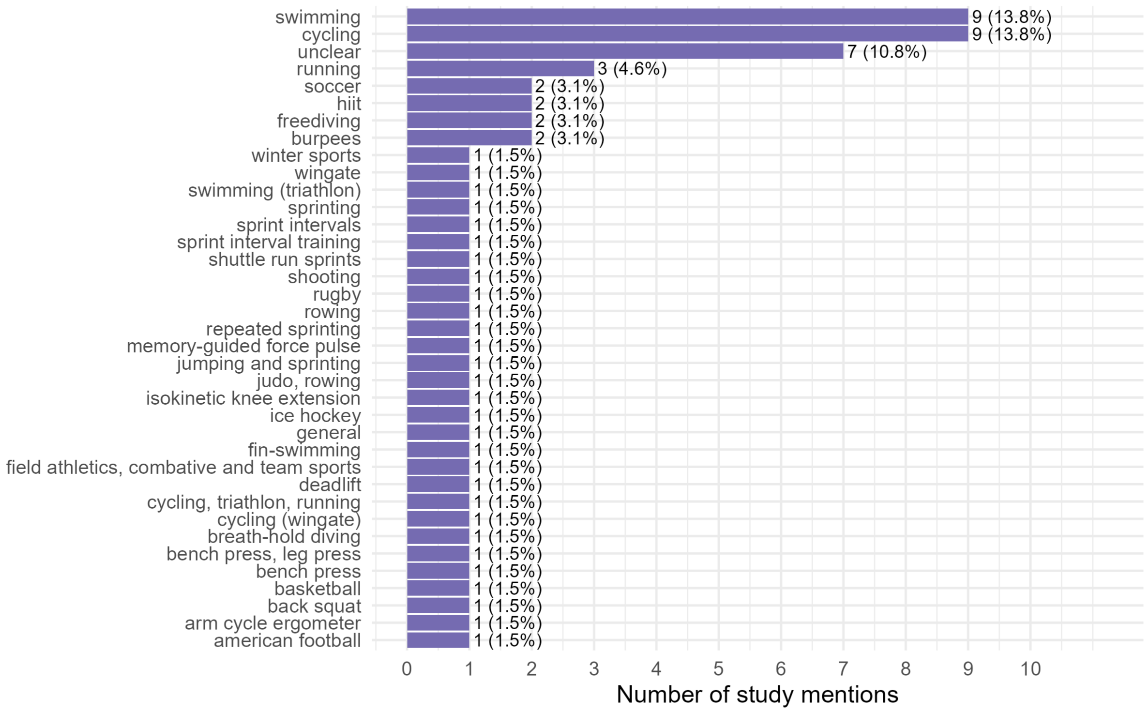


Figure 6: Distribution of sport and exercise contexts.

Figure 6 shows sport-context dispersion. Cycling and swimming were the most represented contexts, but the long tail of single-study sports limits sport-specific certainty and makes cross-sport transfer uncertain.

3.3 Breathing categories and dose

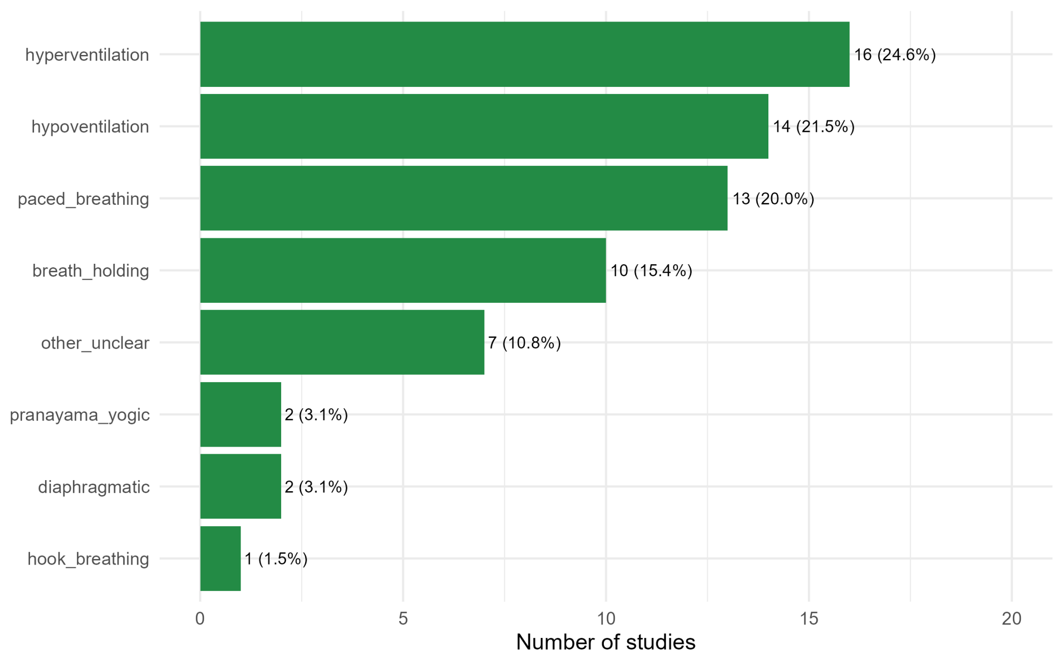


Figure 7: Breathing-category distribution.

Figure 7 reinforces the main conceptual point of the review: breathwork is a family of different interventions. Hyperventilation, hypoventilation, paced breathing, and breath holding dominate the corpus, but their mechanisms and safety profiles differ enough that they should be designed, tested, and interpreted separately.

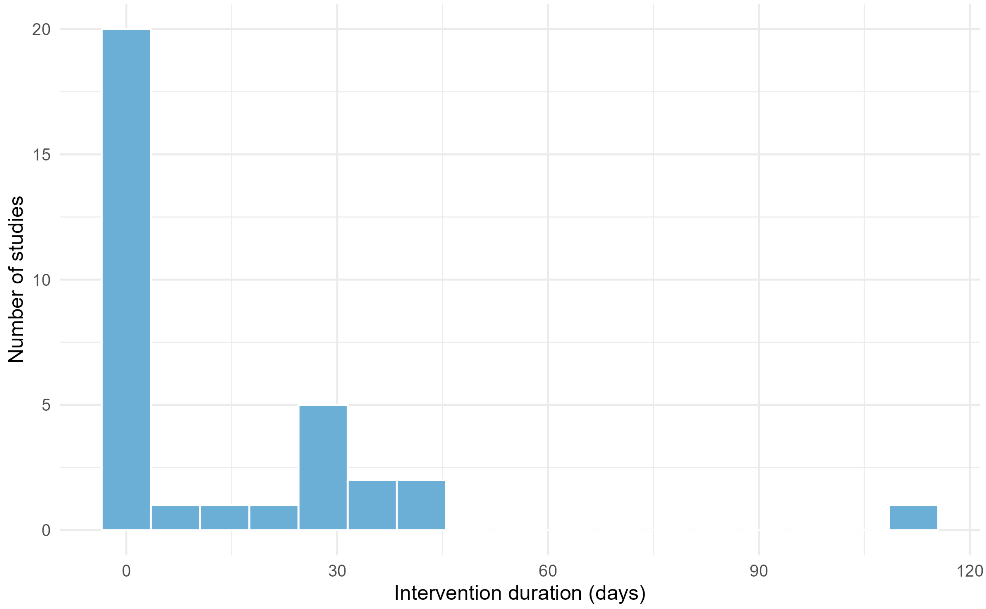


Figure 8: Intervention-duration distribution among reports with parseable duration.

Figure 8 shows that many interventions were acute or short-term. This dose profile explains why the review can say more about immediate physiological or performance responses than about durable training adaptations.

3.4 Efficacy synthesis

Table 3: Exploratory quantitative syntheses.

Synthesis	k	Estimate	95% CI	I^2	Interpretation
Performance efficacy, Hedges' g	20	0.09	-0.16 to 0.34	70.9%	Small pooled direction favoring breathing intervention, but confidence interval crosses the null and prediction interval is wide.
Physiological efficacy, Hedges' g	17	0.21	-0.22 to 0.64	85.9%	Small-to-moderate pooled direction favoring breathing intervention, but uncertainty and heterogeneity are substantial.
Adverse events, risk ratio	29	1.00	0.96 to 1.04	0.0%	No clear overall difference in reported adverse-event risk; sparse-event methods and reporting limitations apply.
Dropout, risk ratio	33	0.91	0.59 to 1.41	0.0%	No clear difference in dropout risk between breathing and comparator conditions.

Table 3 reports the exploratory quantitative syntheses. Performance and physiological outcomes both showed positive pooled directions, but neither confidence interval excluded the null. Heterogeneity was high, especially for physiological outcomes, so the pooled effects are best interpreted as weak signals that justify targeted trials and technique-specific reviews, not as definitive efficacy estimates.

Study / outcome	Weight	SMD [95% CI]
Laborde2019: Number of Burpees	6.7%	-0.11 [-0.47, 0.25]
Devipriya2026: Visual Reaction Time (VRT) Right Hand (ms)	6.5%	1.08 [0.69, 1.46]
Taylor2021: Reaction Time (RT) (ms)	6.4%	0.29 [-0.12, 0.70]
Katz2026: Deadlift 1RM (kg)	6.2%	0.06 [-0.38, 0.50]
Fesseler2025: Cumulative elevation gain (Sprint 1)	6.1%	-0.23 [-0.70, 0.23]
Malakhov2014: Mean velocity of CP displacement (V, mm/s)	5.8%	-1.48 [-1.99, -0.97]
Pelka2017: Average Maximum Speed (m/s)	5.7%	0.03 [-0.50, 0.56]
Tomita2026: Mean Power (%VO2max)	5.7%	0.02 [-0.52, 0.56]
Conlon2022: Shooting Accuracy	5.5%	-0.19 [-0.77, 0.39]
Buxton2024: Average velocity (m/s)	4.9%	0.73 [0.06, 1.41]
Sakamoto2014: Peak Power Pretest (W)	4.7%	-0.03 [-0.75, 0.68]
Sakamoto2015: Peak Torque 60/s (Nm)	4.7%	0.05 [-0.66, 0.77]
Kacoglu2017: 10-m sprint time (s)	4.6%	-0.18 [-0.92, 0.57]
Burtch2017: 200-yard time trial (s)	4.4%	0.14 [-0.65, 0.92]
Deniz2025: Total Volume Lifted (kg)	4.2%	0.56 [-0.26, 1.37]
Robertson2020: 400-m freestyle time (s)	4.0%	0.21 [-0.67, 1.09]
Woorons2019b: RSA mean MPO (W)	3.7%	0.58 [-0.36, 1.53]
Woorons2024: MPOmean(5-8) (W)	3.6%	0.29 [-0.67, 1.25]
Gray2018: 50-yard freestyle time (s)	3.3%	0.14 [-0.91, 1.19]
Lapointe2020: RSASdec (%)	3.1%	0.35 [-0.76, 1.45]

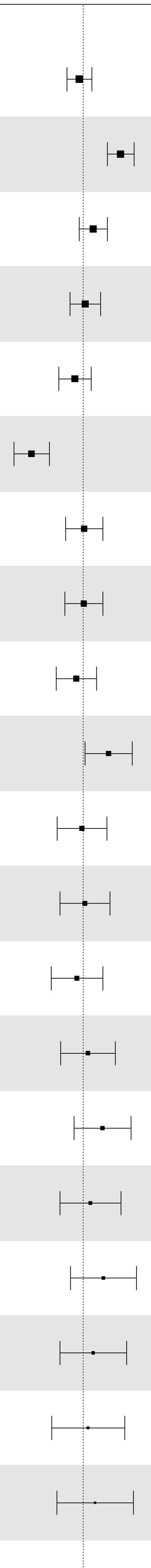
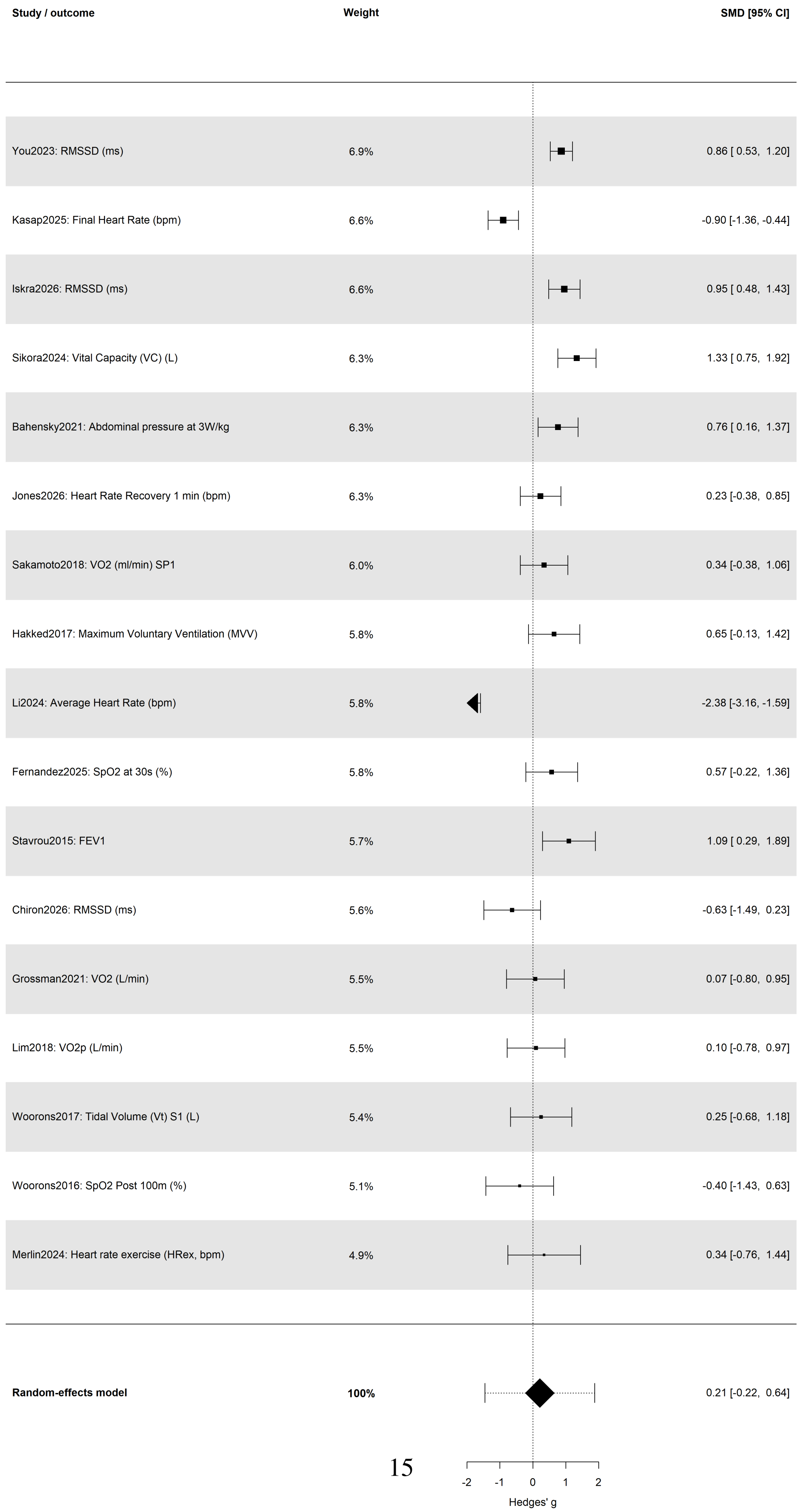


Figure 9 shows a mixed pattern across performance outcomes. Some reports favor the breathing intervention, while others are null or favor the comparator. The wide prediction interval is central to interpretation: a future study could plausibly show benefit, no effect, or harm depending on the technique, sport task, and outcome definition.



Heterogeneity: Q = 114.50, df = 16, p = 0.000; I^2 = 85.9%; tau^2 = 0.675

Figure 10 shows a slightly larger pooled direction than the performance forest plot, but heterogeneity is substantial. Physiological outcomes are mechanistically closer to breathing manipulation, yet they remain diverse: heart-rate, oxygenation, pulmonary-function, autonomic, and perceptual outcomes should not be treated as interchangeable.

3.5 Feasibility and safety

Table 4: Feasibility and safety reporting.

Field	Summary
Adherence	Only 2 reports had parseable adherence values. Adherence was too sparsely and inconsistently reported for meaningful quantitative summary.
Dropout	11 reports with parseable dropout; 871 participants with dropout data; pooled dropout proportion 20.9%.
Adverse-event mentions	Dizziness and tingling appeared in hyperventilation records; pulmonary edema appeared in breath-holding/hook-breathing contexts.
Displayed safety forest plot	Trincat2017 was excluded from the displayed safety risk-ratio plot because the upper confidence interval was extreme (RR upper CI 252.56), making the plot unreadable and uninformative. The study remained part of the extraction record.

Table 4 shows that feasibility and safety reporting remain major weaknesses. Dropout was more extractable than adherence, but adherence was rarely reported in a useful form. Safety reporting was sparse and likely underpowered for uncommon but relevant events, which is a serious limitation for interventions that are often promoted as simple and low risk.

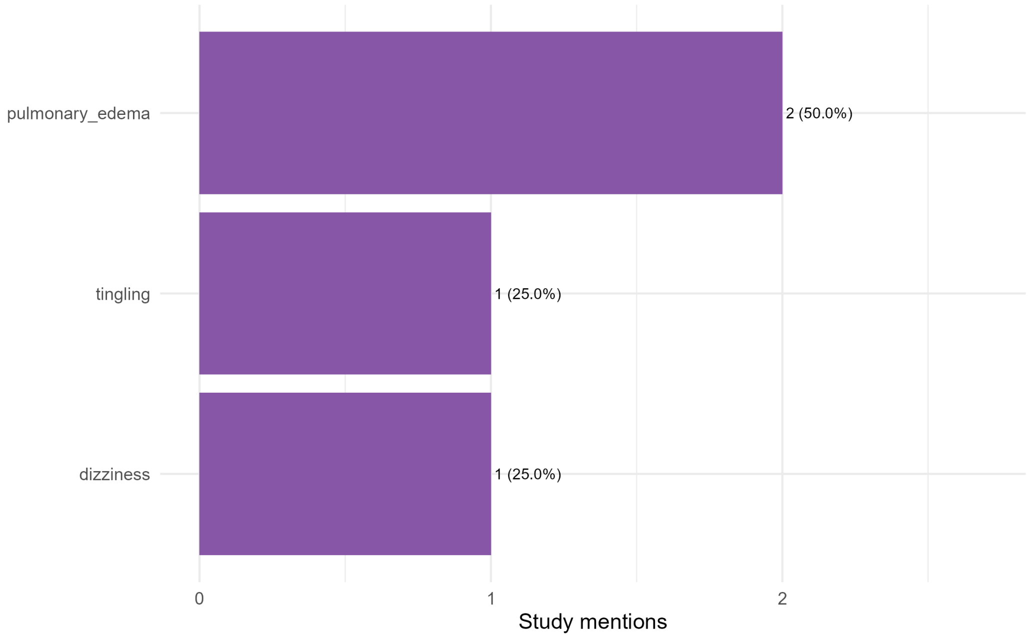


Figure 11: Adverse-event mentions by breathing-technique category.

Figure 11 shows that adverse-event mentions were few. Hyperventilation was associated with sensory symptoms such as dizziness and tingling, while pulmonary edema appeared in breath-holding or hook-breathing contexts. These findings are not incidence estimates, but they identify the technique families where safety monitoring and adverse-event definitions should be strengthened first.

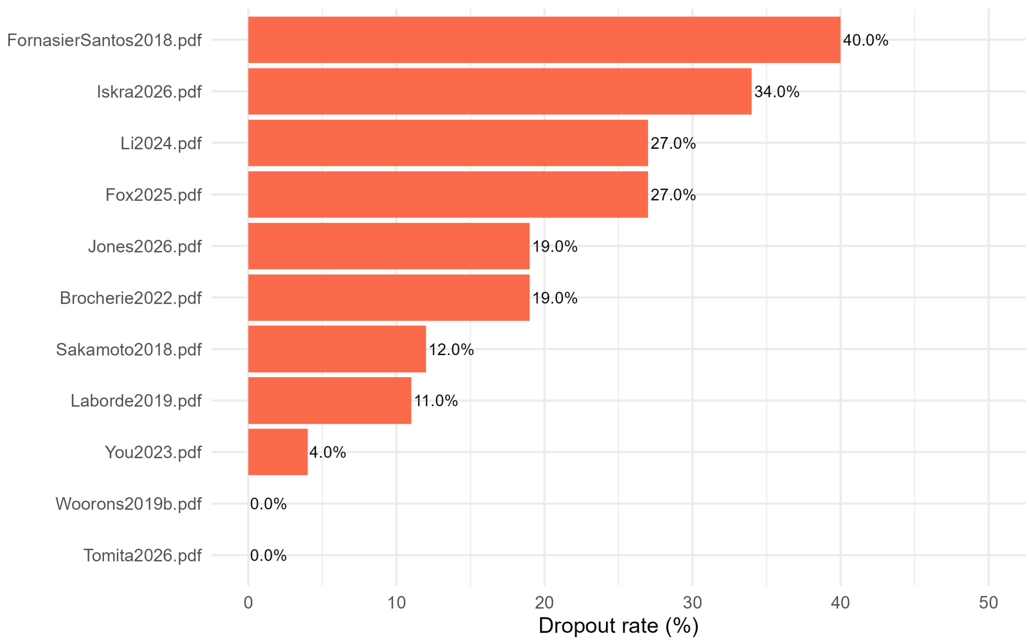


Figure 12: Dropout rates among reports with extractable dropout data.

Figure 12 shows that dropout reporting was available in only a subset of the corpus. The pooled dropout proportion was 20.9% among 871 participants with dropout data, but this should not be generalized to all breathing interventions because missingness is substantial and reasons for discontinuation were not consistently described.

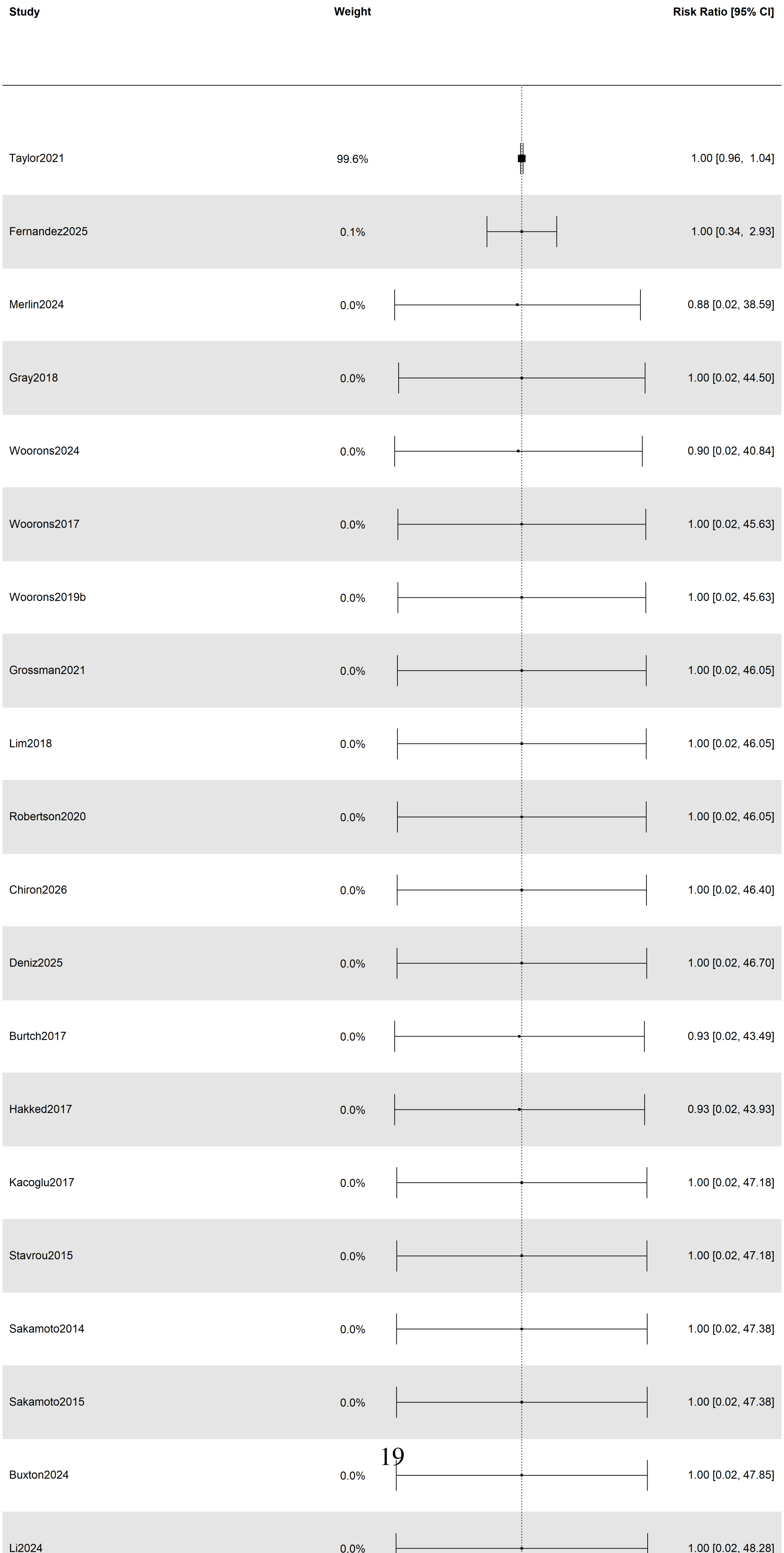

Study
Weight
Risk Ratio [95% CI]
Taylor2021 99.6% 1.00 [0.96, 1.04]
Fernandez2025 0.1% 1.00 [0.34, 2.93]
Merlin2024 0.0% 0.88 [0.02, 38.59]
Gray2018 0.0% 1.00 [0.02, 44.50]
Woorons2024 0.0% 0.90 [0.02, 40.84]
Woorons2017 0.0% 1.00 [0.02, 45.63]
Woorons2019b 0.0% 1.00 [0.02, 45.63]
Grossman2021 0.0% 1.00 [0.02, 46.05]
Lim2018 0.0% 1.00 [0.02, 46.05]
Robertson2020 0.0% 1.00 [0.02, 46.05]
Chiron2026 0.0% 1.00 [0.02, 46.40]
Deniz2025 0.0% 1.00 [0.02, 46.70]
Burtch2017 0.0% 0.93 [0.02, 43.49]
Hakked2017 0.0% 0.93 [0.02, 43.93]
Kacoglu2017 0.0% 1.00 [0.02, 47.18]
Stavrou2015 0.0% 1.00 [0.02, 47.18]
Sakamoto2014 0.0% 1.00 [0.02, 47.38]
Sakamoto2015 0.0% 1.00 [0.02, 47.38]
Buxton2024 0.0% 1.00 [0.02, 47.85]
Li2024 0.0% 1.00 [0.02, 48.28]

Figure 13 shows no clear pooled difference in adverse-event risk between breathing interventions and comparators. This should not be read as proof of safety; the event data are sparse, many studies were not designed primarily to detect adverse events, and rare risks may be invisible in small performance studies.

Study	Weight	Risk Ratio [95% CI]
Li2024	29.0%	1.00 [0.44, 2.25]
You2023	9.1%	1.00 [0.23, 4.26]
Lapointe2020	8.4%	0.90 [0.20, 4.05]
Sakamoto2018	7.2%	1.00 [0.20, 5.09]
Woorons2024	5.6%	0.60 [0.09, 3.79]
Hakked2017	5.3%	0.60 [0.09, 4.02]
Woorons2016	2.0%	0.33 [0.02, 7.14]
Merlin2024	1.3%	0.88 [0.02, 38.59]
Gray2018	1.3%	1.00 [0.02, 44.50]
Trincat2017	1.3%	1.00 [0.02, 45.13]
Woorons2017	1.3%	1.00 [0.02, 45.63]
Woorons2019b	1.3%	1.00 [0.02, 45.63]
Grossman2021	1.3%	1.00 [0.02, 46.05]
Lim2018	1.3%	1.00 [0.02, 46.05]
Robertson2020	1.3%	1.00 [0.02, 46.05]
Chiron2026	1.3%	1.00 [0.02, 46.40]
Deniz2025	1.3%	1.00 [0.02, 46.70]
Burtch2017	1.3%	0.93 [0.02, 43.49]
Fernandez2025	1.3%	1.00 [0.02, 46.96]

Figure 14 shows no clear overall difference in dropout risk. The estimate is compatible with either lower or higher dropout under breathing interventions, so future trials should report adherence, acceptability, and reasons for discontinuation as standard feasibility outcomes.

4 Discussion

4.1 Summary of evidence

This scoping review maps an active but fragmented field. Sixty-five reports were identified across sport, exercise, and performance contexts, with most studies using experimental designs but many relying on crossover, acute, or otherwise heterogeneous protocols. The central finding is not that breathing techniques have a single overall effect. Rather, the evidence base is best understood as a taxonomy and gap map: different breathing families target different mechanisms, are used in different contexts, and require different safety and feasibility considerations.

The exploratory quantitative syntheses support this cautious interpretation. Performance and physiological effects were directionally positive, but confidence intervals crossed the null and heterogeneity was high. These findings are compatible with technique-specific benefit in some settings, no effect in others, and possible context-dependent harms or burdens. The pooled estimates should therefore be read as hypothesis-generating summaries, not as proof that breath-regulation strategies reliably improve sport performance.

4.2 Why technique taxonomy matters

The main conceptual contribution of this review is to keep breathing techniques separated rather than collapsing them into a generic "breathwork" label. This distinction matters physiologically and practically. A technique that changes carbon dioxide tolerance or oxygen saturation is not equivalent to a technique intended to increase vagal activity, reduce arousal, or stabilize attention. Likewise, a single pre-performance breathing routine cannot be interpreted in the same way as a repeated training intervention delivered across weeks.

This review extends prior systematic-review work by mapping study characteristics, athlete level, sport context, dose, feasibility, dropout, and safety alongside efficacy signals. Laborde and colleagues separated breathing techniques in an efficacy-focused synthesis and concluded that heterogeneity, sparse technique-specific evidence, and risk-of-bias concerns limited firm recommendations (Laborde et al., 2024). The present review reaches a compatible conclusion from a broader mapping perspective: the field has enough activity to justify targeted trials, but not enough consistency to support broad practical claims.

4.3 Technique-specific interpretation

Hyperventilation. Hyperventilation protocols were among the most frequent categories. Their sport rationale is usually tied to acute changes in carbon dioxide, pH, arousal, or tolerance of high-intensity work. However, this same mechanism makes safety and tolerability important. Dizziness, tingling, and related symptoms were reported in hyperventilation records, so future studies should treat adverse-event monitoring as part of the intervention protocol rather than as an optional note.

Hypoventilation and breath holding/apnea. Hypoventilation and breath-holding approaches often resemble hypoxic or hypercapnic training strategies, especially when embedded in repeated-sprint or endurance contexts. They may be mechanistically plausible for selected sport tasks, but the evidence remains technique-, dose-, and population-specific. Studies should report breath-hold timing, lung volume, oxygen saturation when relevant, supervision, stopping rules, and symptoms.

Paced, coherent, slow, and diaphragmatic breathing. Paced and slow breathing techniques are more often linked to autonomic regulation, recovery, perceived stress, attentional control, or psychophysiological state. Their effects may be more plausible for recovery and regulation outcomes than for immediate performance gains. The evidence map suggests that future work should avoid treating all slow-breathing protocols as interchangeable: breathing frequency, session timing, biofeedback use, and training duration are likely to matter.

Pranayama/yogic, Wim Hof-type, and hook-breathing approaches. Composite or tradition-derived approaches combine multiple elements, such as controlled hyperventilation, breath holds, postures, meditation, or sport-specific pressure maneuvers. These interventions may be attractive in applied sport settings, but they are difficult to interpret unless the active components are specified. Sport-derived breathing routines should be tested with clear protocols, credible comparators, and transparent reporting of feasibility and safety.

4.4 Feasibility and safety

Feasibility and safety reporting were weaker than efficacy reporting. Adherence was too sparsely and inconsistently reported for meaningful quantitative summary, and dropout reasons were not consistently described. Adverse-event reporting was also sparse. The absence of a clear pooled increase in adverse events should not be interpreted as proof of safety, because many studies were small, not designed for event detection, and inconsistent in reporting. For interventions that alter oxygen saturation, carbon dioxide, intrathoracic pressure, or autonomic state, safety monitoring should be prespecified and reported.

4.5 Implications for coaches and researchers

For practitioners, the evidence supports caution and specificity. Breathing techniques should be matched to the intended mechanism and context rather than adopted as generic breathwork. Slow or paced breathing may be most relevant for regulation, recovery, or attentional control; hypoventilation and breath holding are closer to hypoxic or repeated-sprint training logic; hyperventilation and hook-breathing approaches require particular attention to symptoms and supervision.

For researchers, the next step is not another broad test of "breathwork" but rigorous testing of named techniques. Trials should prespecify breathing rate, depth, breath-hold timing, lung volume, total exposure, comparator, expected mechanism, and outcome domain. They should also report adherence, acceptability, dropout, adverse events, and reasons for discontinuation as standard feasibility outcomes.

4.6 Limitations

Several limitations should be considered. No registered protocol was identified in the available project files, and this should be confirmed before submission. Formal risk-of-bias appraisal was not conducted; this is defensible for a scoping review but limits causal interpretation. Eligibility criteria were broad by design, which allowed the review to map the field but increased heterogeneity across participants, techniques, comparators, timing, dose, and outcomes. Many studies were acute, crossover, or small experimental reports rather than durable training trials. Sample demographics, adherence, dropout, adverse events, and intervention fidelity were incompletely reported. Finally, the pooled Hedges' g and risk-ratio analyses were exploratory and should be interpreted as adjuncts to evidence mapping rather than definitive meta-analytic estimates.

5 Conclusions

Breathing techniques in sport and performance contexts form an active and mechanistically plausible research area, but the evidence is still fragmented. Exploratory performance and physiological pooled effects were positive but uncertain, with confidence intervals crossing the null and substantial heterogeneity. Safety and dropout syntheses did not show clear overall excess risk, but reporting was too incomplete to support strong reassurance. The next phase of research should move from informal sport-derived routines toward clear protocols, credible controls, complete feasibility reporting, prespecified safety monitoring, and mechanism-matched outcomes.

Declarations

Funding: No external funding was declared.
Conflicts of interest: The authors declare no conflicts of interest.
Data availability: The extraction data, generated tables, plots, and scripts are available in the project repository.
Author contributions: The authors contributed to study conception, data charting, analysis, interpretation, and manuscript preparation.

A PRISMA-ScR Checklist Crosswalk

Table 5: PRISMA-ScR checklist crosswalk.

Item	Reporting location	Status
Title and abstract	Title and structured abstract identify the manuscript as a scoping review and summarize source selection, charting, and synthesis.	Reported
Rationale	Introduction explains why heterogeneous breathing strategies require mapping before definitive efficacy recommendations.	Reported
Objectives	Introduction states mapping objectives for study characteristics, interventions, outcomes, feasibility, safety, and exploratory effects.	Reported
Eligibility criteria	Methods describe eligible breathing techniques, populations, and sport/performance contexts.	Reported
Information sources	Methods identify the databases, citation searching, and extraction dataset.	Reported
Selection of sources	Results include Figure 1.	Reported
Data charting	Methods describe bibliometric, participant, intervention, outcome, feasibility, and safety fields.	Reported
Synthesis of results	Results combine descriptive mapping, plots, and exploratory effect synthesis.	Reported
Critical appraisal	Methods state that formal risk-of-bias appraisal was not the central scoping-review objective; reporting completeness is summarized instead.	Reported
Limitations	Discussion describes database/source-set, extraction, reporting, heterogeneity, and meta-analysis limitations.	Reported

Item	Reporting location	Status
Funding and conflicts	Declarations section included.	Reported

Table 5 provides a reporting crosswalk for the main PRISMA-ScR items. It is included to make the review structure transparent and to show where selection, charting, synthesis, limitations, and declarations are addressed in the manuscript.

B Supplementary Study Characteristics

Table S1: Study-level baseline characteristics of the 65 included reports.

Study	Year	Design	Sport/context	Athlete level	N	Age mean (SD)	Male (%)
Durand2000	2000	Obs. cross-sectional	Cycling, Triathlon, Running	elite	22	23.9 (1.3)	100
Spiering2004	2004	Obs. cross-sectional	Winter sports	elite	78	NR	32
Lindholm2005	2005	Obs. cross-sectional	Breath-hold diving	competitive	5	NR	NR
Woorons2008	2008	RCT	Running	competitive	16	27.1 (6.0)	NR
Guimard2014	2014	Acute experimental	Swimming	competitive	15	21.9 (0.9)	100
Malakhov2014	2014	Acute experimental	Field athletics, combative and team sports	competitive	66	19.8 (1.0)	44
Sakamoto2014	2014	Crossover	Cycling	NR	15	NR	NR
Fujii2015	2015	Crossover	Cycling	NR	NR	NR	NR
Jacob2015	2015	Crossover	Swimming	competitive	NR	NR	NR
Sakamoto2015	2015	Crossover	Isokinetic knee extension	competitive	15	21.2 (3.7)	73
Stavrou2015	2015	RCT	Fin-swimming	competitive	28	NR	50
Woorons2016	2016	RCT	Swimming (Triathlon)	competitive	16	NR	75
Burtch2017	2017	RCT	Swimming	competitive	20	NR	NR
Hakked2017	2017	Other experimental	Swimming	competitive	NR	NR	NR
Kacoglu2017	2017	Crossover	Jumping and sprinting	physically active	14	NR	NR
Pelka2017	2017	RCT	Sprinting	competitive	27	25.22 (1.08)	70
Trincat2017	2017	RCT	Swimming	competitive	16	15.55 (1.8)	56
Woorons2017	2017	Crossover	Cycling	competitive	9	27.2 (9.3)	89
FornasierSantos2018	2018	RCT	Rugby	competitive	35	NR	100
Gray2018	2018	Other experimental	Swimming	NR	NR	NR	NR
Lim2018	2018	Crossover	Cycling	physically active	10	24 (3)	100

Continued on next page

Table S1: Study-level baseline characteristics of the 65 included reports (continued).

Study	Year	Design	Sport/context	Athlete level	N	Age mean (SD)	Male (%)
Sakamoto2018	2018	Crossover	Cycling	competitive	17	NR	88
Fernandez2019	2019	Obs. cross-sectional	Freediving	competitive	25	NR	100
Laborde2019	2019	Crossover	Burpees	physically active	120	25.57 (2.52)	58
Woorons2019	2019	Crossover	Shuttle run sprints	competitive	NR	NR	NR
Woorons2019b	2019	RCT	Cycling	competitive	18	34.6 (11)	100
Bouten2020	2020	Crossover	Cycling	physically active	11	NR	NR
Lapointe2020	2020	RCT	Basketball	competitive	17	22.3 (1.2)	71
Robertson2020	2020	Crossover	Swimming	competitive	NR	NR	NR
Sakamoto2020	2020	Crossover	Bench press, leg press	competitive	11	22.5 (4.3)	100
Woorons2020	2020	NR	NR	NR	NR	NR	NR
Bahensky2021	2021	RCT	NR	NR	NR	NR	NR
Citherlet2021	2021	Crossover	Repeated sprinting	competitive	NR	NR	NR
Dobashi2021	2021	Crossover	Cycling (Wingate)	competitive	11	24 (2)	91
Grossman2021	2021	Crossover	Cycling	physically active	10	23.7 (2.5)	100
Taylor2021	2021	Crossover	Memory-guided force pulse	healthy non-athlete	50	21.3 (2.8)	36
Vaidya2021	2021	NR	NR	NR	NR	NR	NR
Brocherie2022	2022	RCT	Ice hockey	elite	43	16.7 (1.5)	100
Conlon2022	2022	RCT	Shooting	competitive	64	NR	NR
Marko2022	2022	Before-after	Cycling	physically active	NR	NR	NR
You2023	2023	Crossover	NA	physically active	78	22.32 (2.11)	71
Braham2024	2024	RCT	Soccer	competitive	NR	NR	NR

Continued on next page

Table S1: Study-level baseline characteristics of the 65 included reports (continued).

Study	Year	Design	Sport/context	Athlete level	N	Age mean (SD)	Male (%)
Buxton2024	2024	Crossover	Back Squat	recreationally trained	18	NR	100
Li2024	2024	Crossover	Burpees	physically active	30	NR	NR
Merlin2024	2024	RCT	Swimming	elite	13	13.9 (0.38)	38
Rosa2024	2024	Crossover	Arm cycle ergometer	NR	14	NR	NR
Sikora2024	2024	Obs. cross-sectional	NR	competitive	69	NR	NR
Woorons2024	2024	RCT	Judo, Rowing	elite	20	20.6 (2.5)	50
Deniz2025	2025	Crossover	Bench press	recreationally trained	12	27.92 (7.38)	100
Fernandez2025	2025	Crossover	Freediving	competitive	13	42 (9)	NR
Fesseler2025	2025	Crossover	Sprint intervals	recreationally trained	36	NR	NR
Fox2025	2025	RCT	General	physically active	404	37 (9.95)	44
Kasap2025	2025	Crossover	HIIT	competitive	40	NR	NR
Liapaki2025	2025	Crossover	Running	competitive	NR	24.7 (3.1)	NR
Lorinczi2025	2025	RCT	Soccer	competitive	NR	NR	NR
Yilmaz2025	2025	Other experimental	Wingate	physically active	NR	NR	NR
Chiron2026	2026	RCT	Running	elite	22	22.8 (3.3)	68
Devipriya2026	2026	Before-after	NA	healthy non-athlete	60	NR	50
FernandezBarradas2026	2026	Before-after	American football	competitive	NR	NR	NR
Iskra2026	2026	Crossover	NA	physically active	38	NR	NR

Continued on next page

Table S1: Study-level baseline characteristics of the 65 included reports (continued).

Study	Year	Design	Sport/context	Athlete level	N	Age mean (SD)	Male (%)
Jing2026	2026	RCT	Swimming	competitive	NR	NR	NR
Jones2026	2026	RCT	Rowing	physically active	62	21.0 (3.0)	70
Katz2026	2026	Other experimental	Deadlift	physically active	40	NR	NR
Raidl2026	2026	RCT	Sprint interval training	competitive	NR	NR	NR
Tomita2026	2026	Crossover	HIIT	physically active	26	NR	NR

Note. NR = not reported; Obs. = observational. Values are shown as reported or extractable from the source papers.

Supplementary Table S1 provides the study-level extraction rows behind the aggregate summaries in the main text. It is placed in the appendix because the 65-row format is useful for transparency but too large for the main Results flow.

C Exact Search Strings

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    OR "Athletes"[Mesh]
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    OR "controlled frequency breathing"[Title/Abstract]
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    OR "coherent breathing"[Title/Abstract]
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    OR "resonance frequency breathing"[Title/Abstract]
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    OR "slow-paced breathing"[Title/Abstract]
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